

AeroForge VR: Design-to-Flight

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Figure 1: AeroForge VR overview. Participants use Quest 3 to author a primitive aircraft blockout (B–C), view its SDF representation (D), receive the generated aircraft mesh in the headset (E), and continue with flight and terrain interaction in the cloudscape (A).

ABSTRACT

AeroForge VR is designed as an entry point to 3D authoring for participants with limited experience using conventional 3D modeling tools. Participants assemble and transform primitives to express an aircraft’s coarse spatial structure and proportions. The system encodes the blockout as a signed distance field (SDF) and retrieves airplane priors. Both condition a 3D diffusion model. After generation, participants navigate their aircraft through a cloudscape and paint mountain terrain. They experience their generated aircraft in motion and explore its spatial form from multiple viewpoints. This design-to-flight workflow integrates spatial authoring, structure-conditioned generation, and in-headset interaction into an immersive creative experience.

Index Terms: Virtual reality, interactive systems and tools, mesh geometry models.

1 RELATED WORK

SDF-Based Conditional 3D Generation. Recent advances in 3D generation have enabled shape synthesis and completion from partial geometric inputs. Among approaches that use explicit geometric conditioning, SDFusion and Diffusion-SDF employ signed-distance representations for conditional 3D generation and shape completion [1, 2]. These works show how SDFs can represent geometric structure as a condition for 3D synthesis.

Multi-View Depth-Map Completion. View-based methods represent 3D shapes through depth images rendered from fixed viewpoints. Render4Completion completes multi-view depth maps in the 2D image domain and back-projects them to recover 3D geometry [4]. This formulation motivates AeroForge’s use of canonical depth views for refining generated geometry.

Immersive and Proxy-Guided 3D Authoring. VRCopilot integrates pretrained generative models and intermediate representa-

tions into a mixed-initiative VR workflow for 3D layout authoring [7]. Coin3D uses coarse geometry proxies assembled from basic shapes to guide interactive 3D asset generation [3]. These systems illustrate complementary roles for immersive interaction and explicit geometric guidance in generative 3D authoring.

AeroForge combines these directions in a design-to-flight XR workflow. A participant-authored blockout provides the input SDF for airplane retrieval and diffusion conditioning. Canonical depth views refine the result. The aircraft returns to the same XR application for flight and terrain interaction.

2 AEROFORGE VR SYSTEM

AeroForge consists of two connected stages implemented in Unity within a single Meta Quest 3 application. In the first, participants assemble a primitive aircraft blockout. A generation backend processes the blockout and returns an aircraft mesh to the application. In the second, participants navigate the generated aircraft through a cloudscape and paint mountain terrain. Figure 2 summarizes the aircraft-generation pipeline.

2.1 Design Rationale

AeroForge focuses participant input on component relationships and proportions, while the generation pipeline handles detailed geometric completion. This authoring model is intended for participants with limited conventional 3D modeling experience because it does not require them to construct a complete aircraft mesh manually. After generation, the second stage presents the generated aircraft in a dynamic flight setting. Participants navigate it to experience its spatial form in motion and from changing viewpoints. Terrain painting extends authoring to the surrounding environment, allowing participants to shape the setting in which they experience the aircraft. XR unifies blockout manipulation, *SDF Preview*, aircraft navigation, and terrain painting in one spatial context.

2.2 Aircraft Design and Generation

The authoring stage provides a VR workspace with basic shape and transformation controls. Participants add boxes, cylinders, and spheres, then translate, rotate, stretch, or compress them with hand-held controllers. The resulting blockout records component place-

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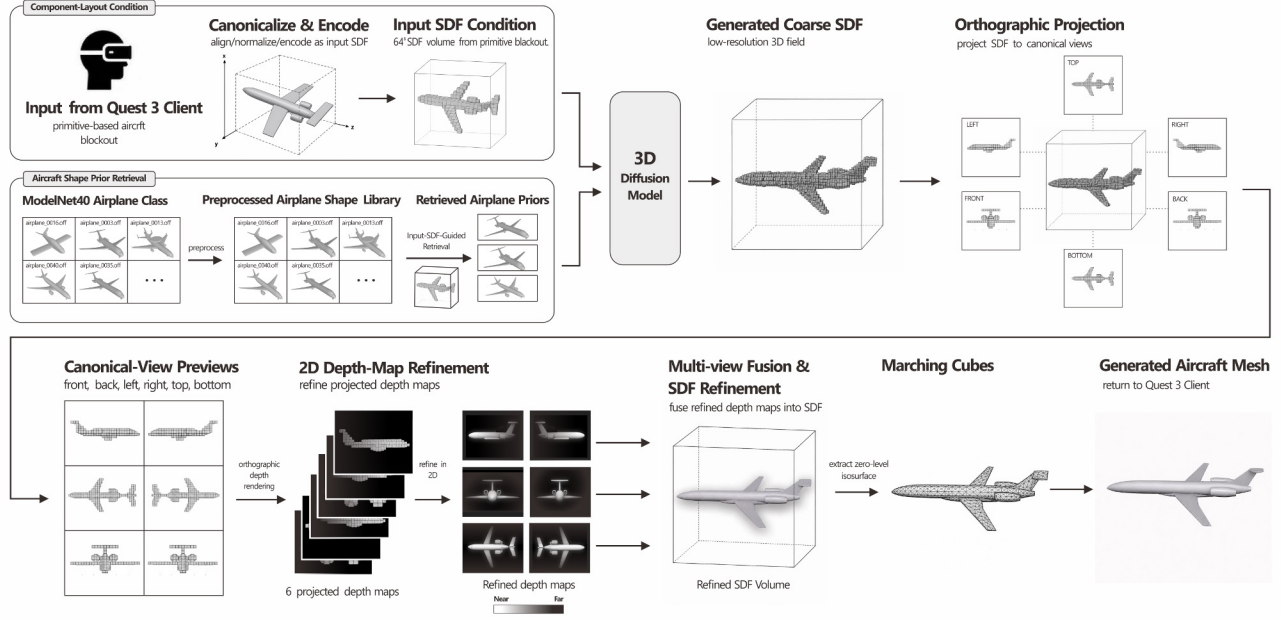


Figure 2: Overview of the AeroForge VR aircraft-generation pipeline. The participant’s primitive blockout is encoded as an input SDF, which, together with retrieved airplane priors, conditions a 3D diffusion model. The resulting coarse SDF is projected into canonical-view depth maps and refined in 2D. The refined maps are fused into a refined SDF volume, which is converted into a mesh using Marching Cubes and returned to the Quest 3 client.

ment, relative scale, and overall aircraft proportion, and serves as the structural input to generation.

After completing the blockout, participants activate *SDF Preview* to view the SDF representation derived from the blockout. The system canonicalizes the primitive assembly and samples it on a $64 \times 64 \times 64$ grid to construct an input SDF. The resulting input SDF combines the separate primitives into a single volumetric representation in a common coordinate frame. The same SDF serves both as the query for retrieving similar airplane shapes and as the component-layout condition for generation.

The system compares the input SDF with a preprocessed shape library derived from the airplane class of ModelNet40 [6] and retrieves similar airplane shapes as priors for generation. The input SDF and the retrieved priors jointly condition a 3D diffusion model, which produces a coarse aircraft SDF.

During generation, the coarse SDF is orthographically rendered from six canonical viewpoints to produce projected depth maps. A 2D depth-map refinement stage produces one refined depth map for each view. The refined maps are fused into a refined SDF volume. Marching Cubes extracts the zero-level isosurface as a triangle mesh [5]. The resulting mesh is returned to the Quest 3 application as the participant’s generated aircraft.

2.3 In-Headset Flight and Terrain Interaction

After generation, participants enter a stylized virtual flight experience set in a cloudscape inspired by Chinese *shan-shui* painting. They guide the generated aircraft with the right-hand trigger, while atmospheric motion and calm music establish the audiovisual setting. All generated aircraft use consistent navigation controls. Participants experience the aircraft in motion and view its overall spatial form from changing viewpoints.

The left-hand trigger acts as an ink brush for painting mountain terrain. Participants shape the environment while navigating their aircraft, extending the creative process from object authoring to environmental authoring. Painted terrain forms interactive spatial fea-

tures that can redirect the aircraft on contact.

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